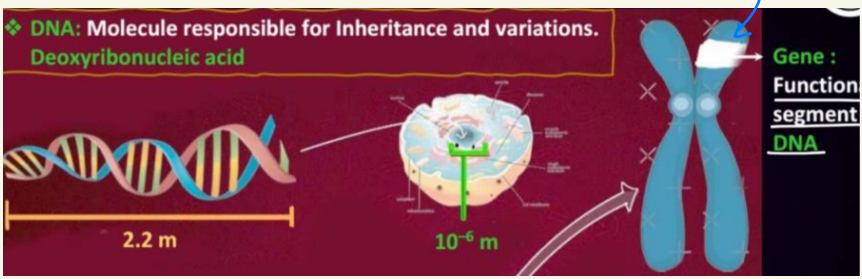
12th hacker series

Principles of inheritance and variation

Genetics

- genetics- combined study of inheritance and variations
- Inheritance- process of transfer of characters from parents to offspring
- variation- degree by which progeny differ from their parents
- Gregor Mendel- father of genetics
- 8000- 1000 BC- cause of variation was hidden in sexual reproduction
- Artificial selection- Indian breed— Sahiwal cow in Punjab
- little idea about scientific basis

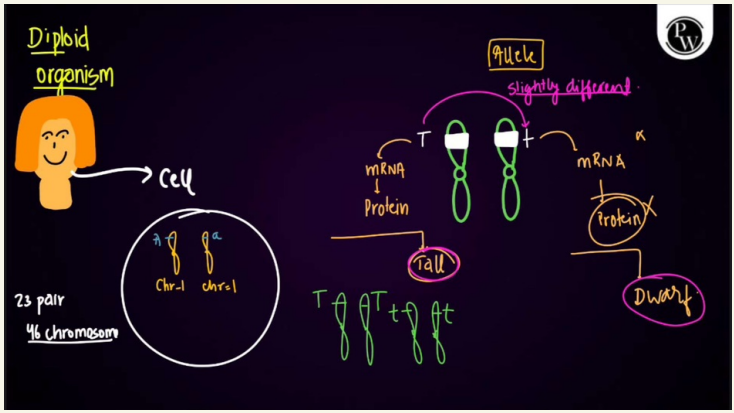
DNA- molecule, responsible for inheritance and variation (deoxyribonucleic acid)
Packaging is done
Maximum condensed form is called as chromosome



- locus: physical location of gene on chromosome
- Character: features of organism. For example.- plant height ,flower colour.
- allele: genes which code for a pair of contrasting traits slightly different forms of same gene.
- Homologous chromosome: same locus of every gene



- trait: specific characterisation. For example.- tall, dwarf.



- homozygous allele: pure line/ true breeding
  Similar allele in diploid cell
  TT or tt
- heterozygous allele
  Dissimilar alleles in diploid cell
  Tt
- dominant AlLele: Allele which express itself even in heterozygous condition
- Recessive allele: which express itself only in homozygous condition
- Phenotype: observable character, for example- tall, dwarf
- Genotype: genetic composition (TT/ tt/ Tt)

Aa

Types of gamete- 2

n- 1

$2^1 = 2 \rightarrow A, a$

AA

Types of gamete- 1

n- 0

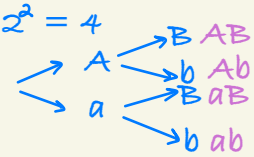
$2^0 = 1 \rightarrow A$

AaBb

Types of gamete- 4

n- 2

$2^2 = 4$

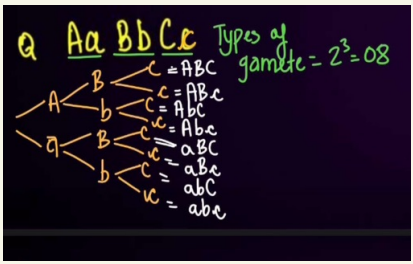


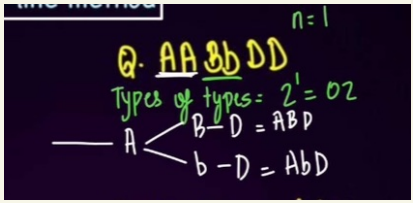
Fork line method



Homozygous

Heterozygous





Conducted hybridisation experiment

- 1856- 1863 (7 years)
- Proposed law of inheritance in living organisms
- after 16 years of his death, Carl correns Hugo de vries and Von Tschermak- rediscovered work independently

Why Mendel's experiment were successful?

- statistical analysis in maths logic
- Large sampling size. More credibility.
- Use of garden pea plant

- experiments on successive generation. ⟶ general rule of inheritance.

Reason for unrecognised work

- No physical proof for existence of factor
- Communication was not easy as it is now
- Use of maths logic to explain biology was new and unacceptable for the biologist at that time
- His concept of gene as stable discrete units, which did not blend was not acceptable by his contemporary

Pea plants
characters = 07

Contrasting trait = 14.

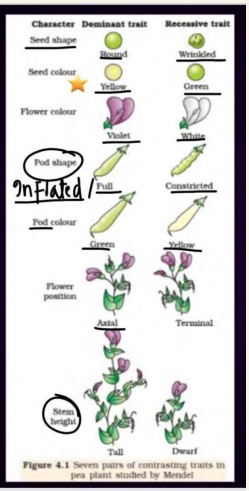


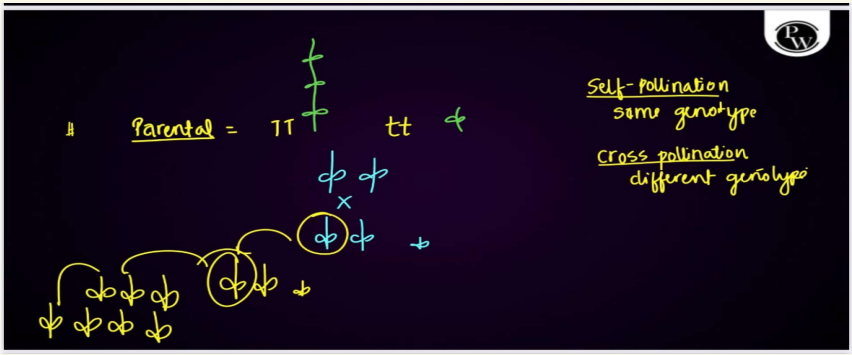
Figure 4.1 Seven pairs of contrasting traits in pea plant studied by Mendel

Why Mendel used garden Pea

- Short life cycle
- Easy to grow
- Naturally self pollination hybridisation (Artificial)
- Contrasting traits
- Produce large number of seeds
- Bisexual flowers

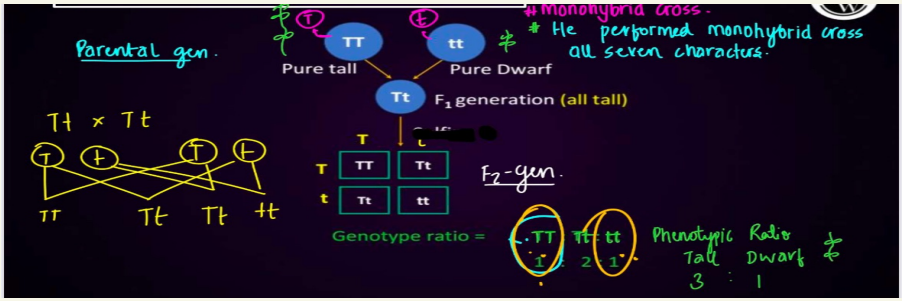
Mendel conducted cross pollination (Artificially) using true breeding Pea line

- He selected 14 true breeding contrasting traits
- Undergoes continuous self pollination and shows stable trait inheritance and expression for several genes



Inheritance of one gene

- Single character is considered
- Monohybrid cross
- he performed monohybrid cross all seven characters



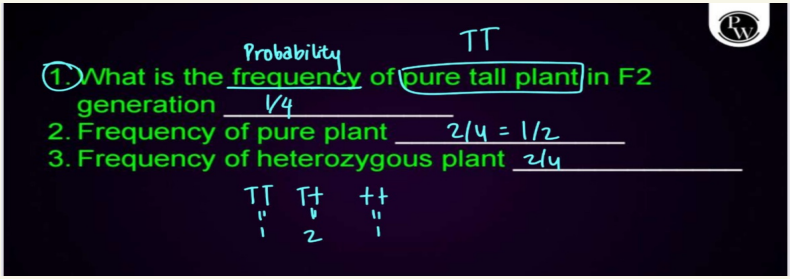
Formula for F2 generation

Types of phenotype $= 2^n$

Types of genotype $= 3^n$

Zygotic combination $= 4^n$

n= number of heterozygous pair



Law of dominance (first law of inheritance)

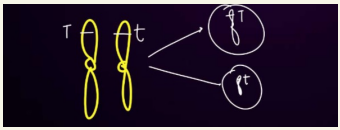
- Characters are controlled by discrete units called factors/today known as gene
- Factors occur in pair
- In case of dissimilar factors → One member of the pair dominate the other.
- This explains that only one of the parental characters expressed in F1
- Proportion of 3:1 obtained in F2

Exception of law of dominance

- Incomplete dominance
- Co-dominance

Law of segregation/ law of purity of gametes

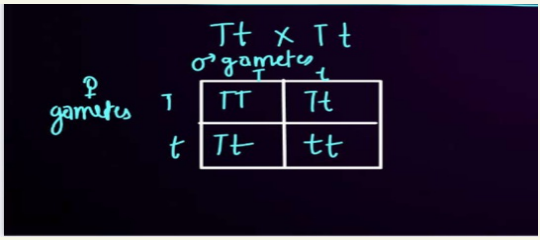
- Based on the fact that alleles do not show, blending and both characters are recovered at such in F2 generation
- Pair of factors or allele segregate from each other during gamete formation and gamete receive only one of the two factors (universal law)



Punnet Square

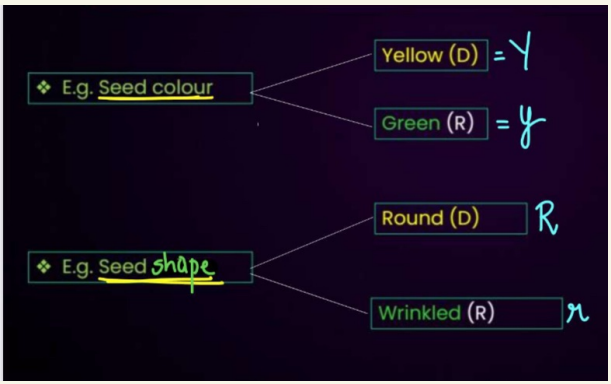
→ NEET 2019 PYQ

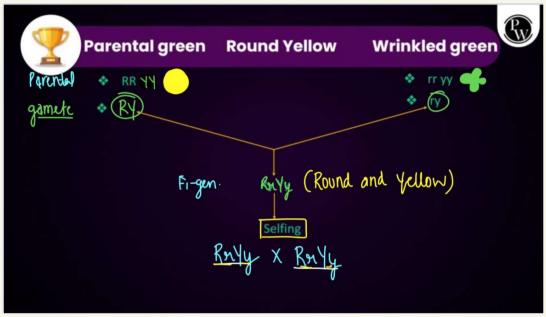
- developed by British geneticist - Reginald C punnet
- Graphical representation to calculate the probability of all possible genotype of offspring

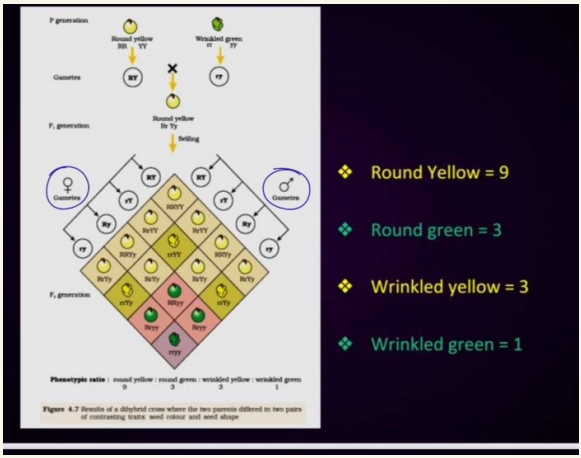


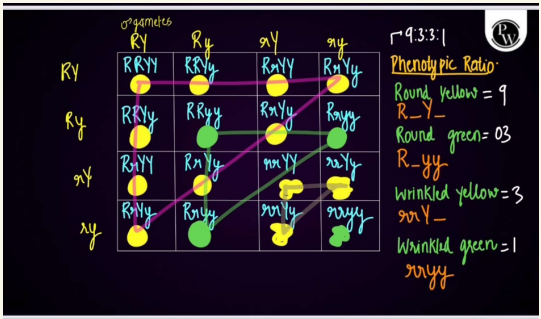
Inheritance of two genes/Dihybrid cross

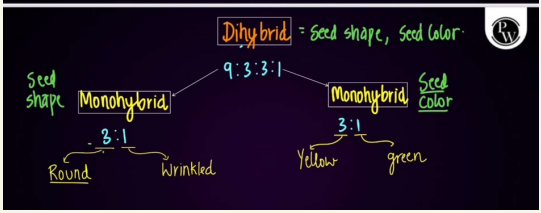
- Two characters are considered
- to study the impact of inheritance of one character over the other





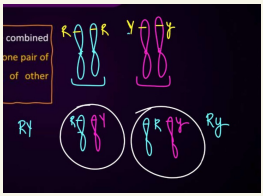






Law of independent assortment

- When two pair of traits are considered in a hybrid, segregation of one pair of character is independent of the other pair of character
- Exception- linkage



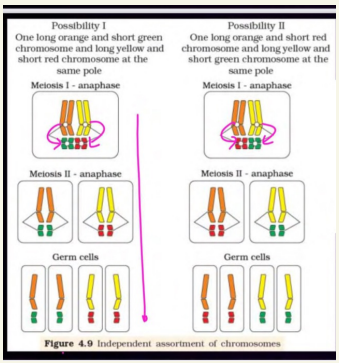


Figure 4.9 Independent assortment of chromosomes

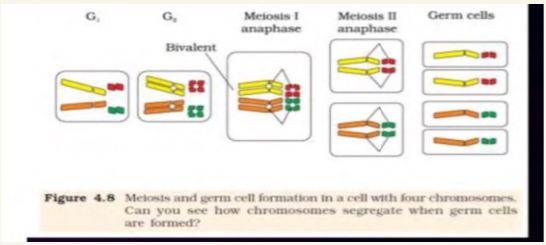
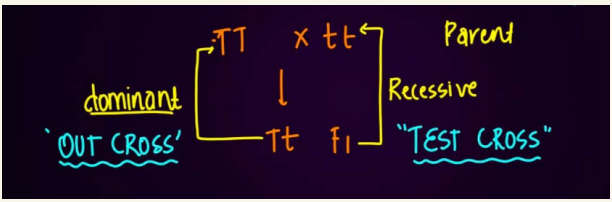


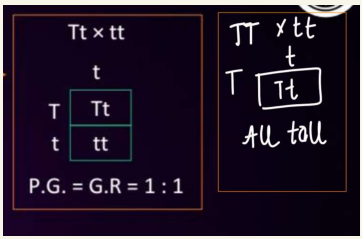
Figure 4.8 Meiosis and germ cell formation in a cell with four chromosomes. Can you see how chromosomes segregate when germ cells are formed?

Special cross

- back cross- cross with parental gene
  - Dominant
  - Recessive

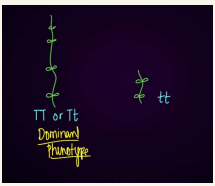


- test cross- Uses-
  - to predict genotype
  - to know types of gametes formed



F1 generation/F2 Cross with recessive parent

Plan showing dominant trait is crossed with recessive parent

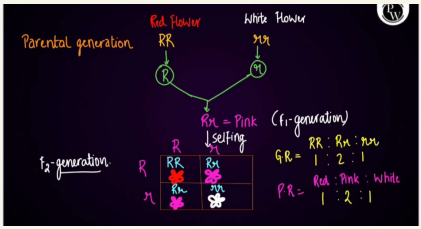


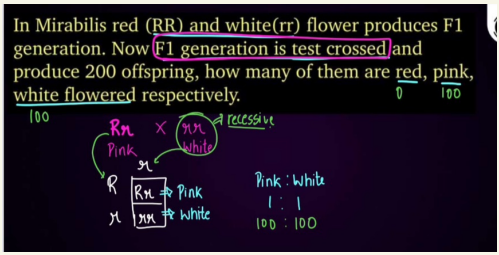
Incomplete dominance

- Exception of first law of inheritance
- Dominant trait is not expressed completely
- F1- matches neither of the parent
- Example- dog flower(snapdragon or antirrhinum Manjus, Mirabilis Jalapa (4 o'clock plant)
- PR= GR
- Due to Modified allele all could be responsible for
  1. normal/less efficient enzyme
  2. Non-functional enzyme
  3. no enzyme at all



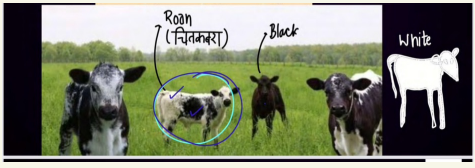






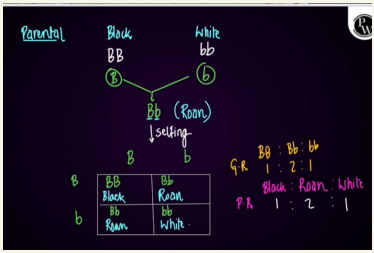
Codominance

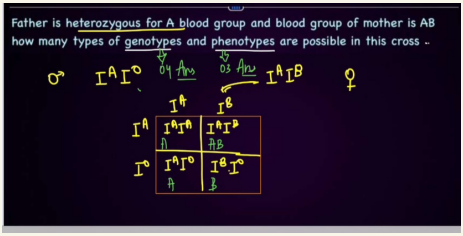
- exception of law dominance
- in heterozygous condition, both alleles express them equally
- F1 generation- matches either of the parents
- P R= GR
- Example- AB blood group
  Coat colour in short horned cattle



Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood types of offspring
I^A	I^A	I^AI^A	A
I^A	I^B	I^AI^B	AB
I^A	i	I^Ai	A
I^B	I^A	I^AI^B	AB
I^B	I^B	I^BI^B	B
I^B	i	I^Bi	B
i	i	ii	O

Co-dominance.



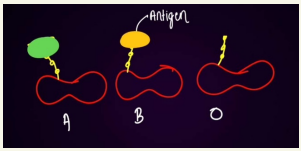


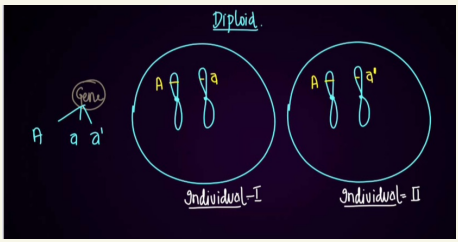
A child with blood group O his parent's blood group can't be

1. B and O
2. A and O
3. AB
4. A and B

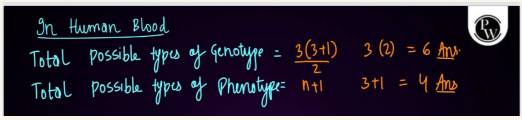
Multiple allelism

- presence of more than two alleles for a gene
- Arise due to mutation
- Studied at population level
- Types of genotype= $n(n+1)/2$
- Types of phenotype= $n+1$
- Example- ABO blood group
- Regulated by I gene, has 3 alleles — I^A, I^B, i
- polymer of sugar that protrude out of the cell membrane is called as antigen which decides blood group
- I-gene decides antigen.⟶Polymer of sugar.









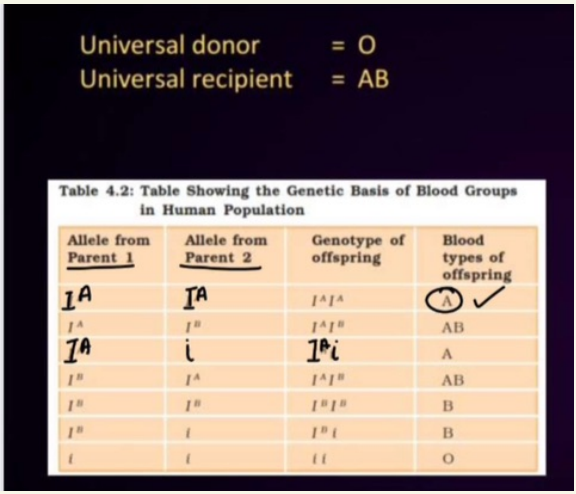
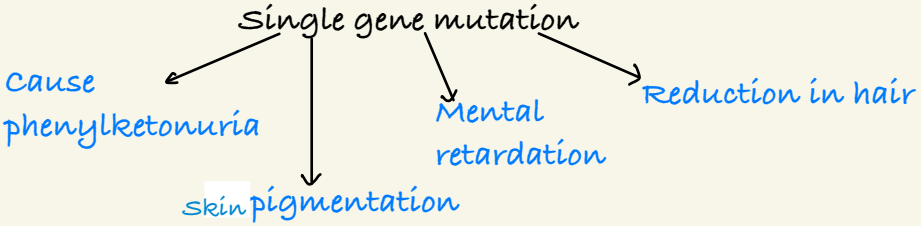


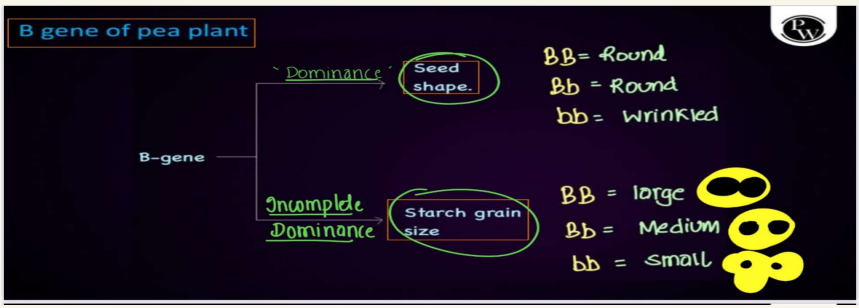
Table 4.2: Table Showing the Genetic Basis of Blood Groups in Human Population

Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood types of offspring
I^A	I^A	I^AI^A	A ✓
I^A	I^B	I^AI^B	AB
I^A	i	I^Ai	A
I^B	I^A	I^AI^B	AB
I^B	I^B	I^BI^B	B
I^B	i	I^Bi	B
i	i	ii	O

Pleiotropy

- Single gene exhibit multiple phenotypic expression
- Because product of gene is involved in different metabolic pathway
- Example- phenylketonuria





- dominance is not an autonomous feature of gene
- it depends on the product of particular trait we choose.



Three or more gene control a character

Trait depends on number of dominant allele

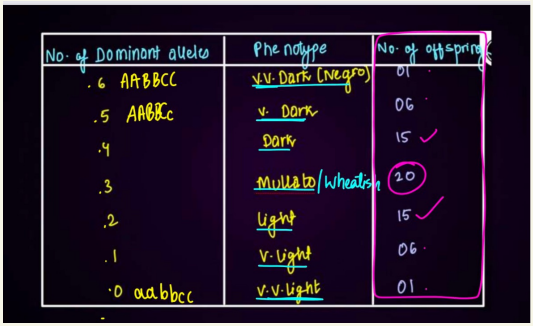
Effect of each allele is additive

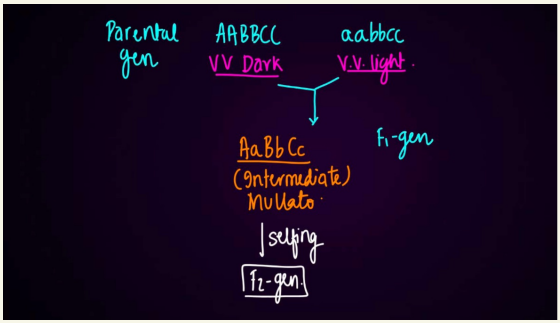
Influence of environment

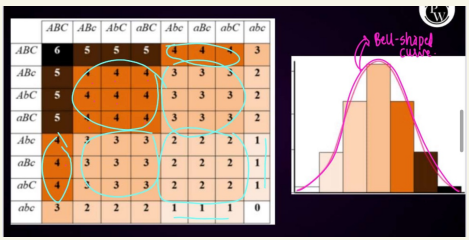
Polygenic inheritance

- assume= human skin colour regulated by three gene A, B, C
- AA BB CC= darkest
- aa bb cc= Lightest
- Aa Bb Cc = intermediate

Example-
- human skin colour
- Human height

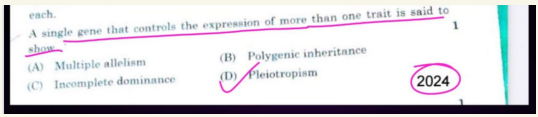








- types of phenotype= 2n+ 1
- Types of genotype= 3^n
- Zygotic combination= 4^n



Chromosomal theory of inheritance

- in 1902, Walter Sutton, Theodore boveri noted chromosomal behaviour is parallel to the factor which was explained by Mendel
- Sutton united the knowledge of chromosome segregation with Mendelian principal and called it chromosomal theory of inheritance

Table 4.3: A Comparison between the Behaviour of Chromosomes and Genes

A = Gene	B = Chromo me
Occur in pairs	Occur in pairs
Segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete	Segregate at gamete formation and only one of each pair is transmitted to a gamete
Independent pairs segregate independently of each other	One pair segregates independently of another pair
Can you tell which of these columns A or B represent the chromosome and which represents the gene? How did you decide?	

Thomas Hunt Morgan

- experimental verification of chromosome theory of inheritance was given by him and his colleagues
- father of experimental genetics
- he worked on fruit Fly- Drosophila melanogaster



Why worked on fruit fly?

- short life cycle (approx two weeks)
- Easily grown in simple synthetic medium
- Produce large number of progeny in single mating
- Male and female easily distinguishable
- Hereditary characters easily observed under low power microscope
- 2n=8
  - 6-Autosome
  - 2- sex chromosome

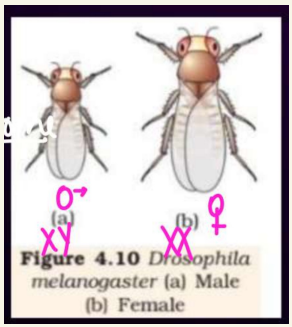
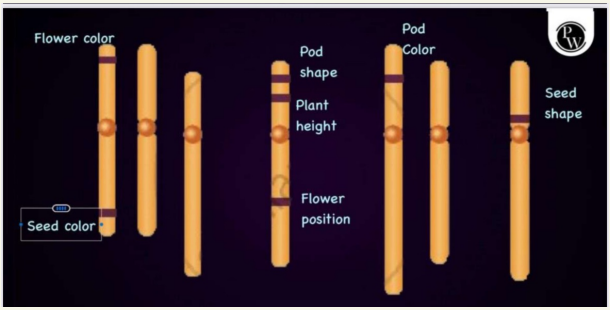


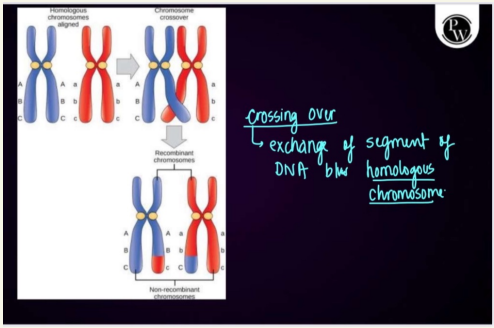
Figure 4.10 *Drosophila melanogaster* (a) Male (b) Female

- he performed dihybrid cross
- F2 ratio derived from 9:3:3:1
- Due to linkage
- Physical association of genes on chromosome

Linkage

- Exception of law of independent assortment
- Physical association of genes on chromosome
- Term linkage was given by TH Morgan
- Linked genes are present on same chromosome at different locus, but can be separated by crossing over
- More distance, more crossing over, less linkage
- Less distance, less crossing over, more linkage
- Crossing over. ⟶ 1/linkage.



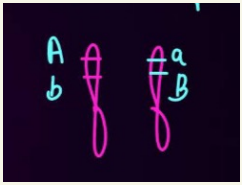


arrangement of linked genes

CIS/coupling



Trans/ repulsion



Types of linked genes

- loosely linked
  - linkage less
  - Recombination more
  - Distance more
- tightly linked
  - linkage more
  - Recombination less
  - Distance less

Linkage

Complete linkage

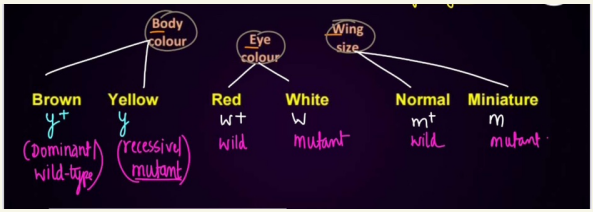
- crossing over does not occurs between genes
- very rare
- Recombinant= 0
- Parental type= 100%

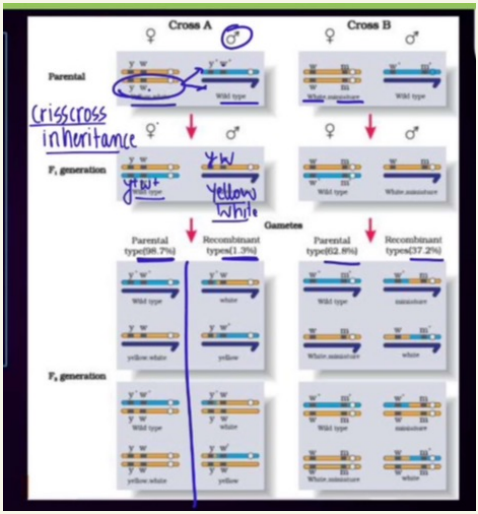
Incomplete linkage

- crossing over occurs between genes
- Common
- Recombinants are formed
- Parental type> New combination

Three characters considered by Morgan

- Genes of all characters are present on X chromosome



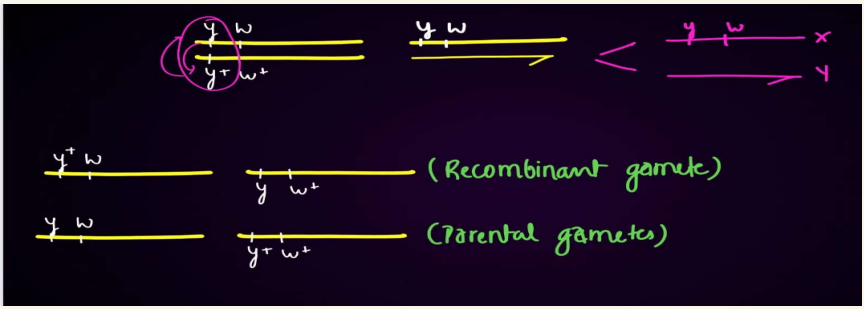


Cross A.

- body colour
- Eye colour
- Recombination less
- Tightly linked genes
- %R= 1.3%
- %P= 98.7%

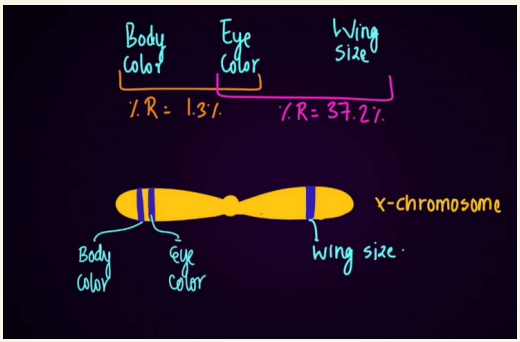
Cross B

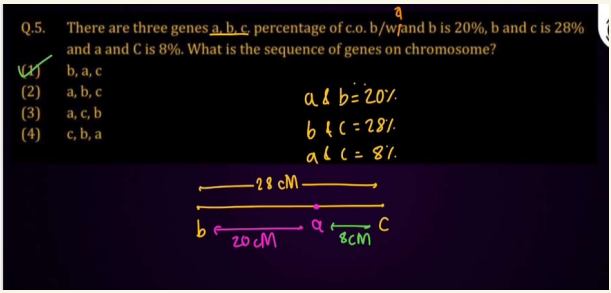
- eye colour
- Wing size
- Recombination more
- Loosely linked genes
- %R= 37.2%
- %P= 62.8%



Genetic map

- representation of position of linked genes on a chromosome
- First given by Morgan's student, Alfred sturtevent
- % recombination = $\frac{\text{number of recombinant.}}{\text{total number of progeny}} \times 100$
- 1% of recombination=1 map unit/ centimorgan cM





Sex determination

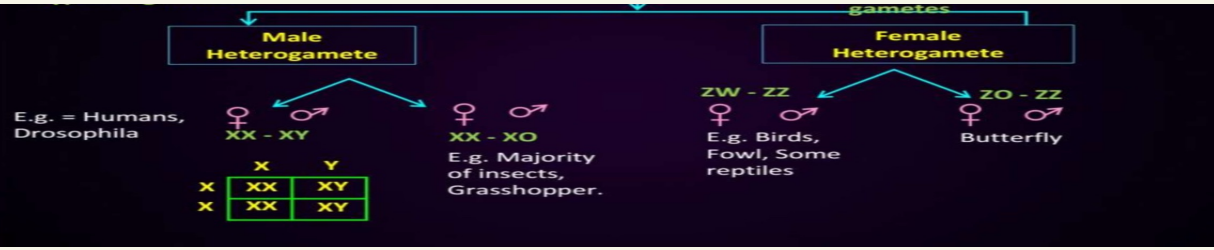
- Henking in 1891
- Observed spermatogenesis of few insects
- Observed nucleur structure of sperm- 50% X body (+) present
- 50% = X body (-) present
- Later X body was found to be chromosome (X chromosome)

Male heterogamete

- sex determination depends on male
- produce different types of gametes

Female heterogamete

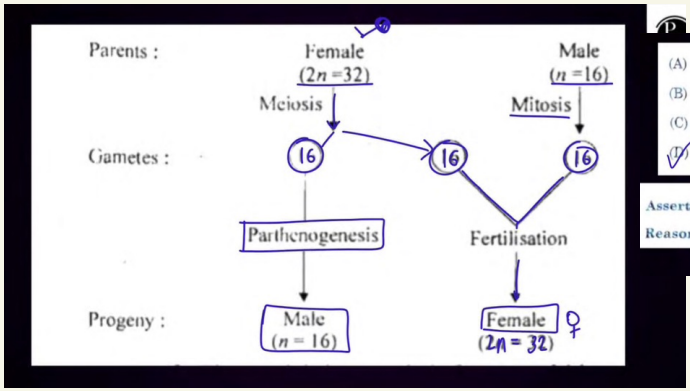
- sex determination depends on female
- Female produce different types of gametes



Haploid diploid mechanism

- number of set of chromosome responsible
- Example- honeybee
  n = drone (male)
  2n = female → Worker Bee
  → Queen

- Drones do not have father, cannot have son, but have grandfather and can have grandson



(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true. 2024

Assertion (A) : In birds the sex of the offspring is determined by male. 1
Reason (R) : Males are homogametic while females are heterogametic.

Mutation

Alteration of DNA that results in change in genotype and phenotype

Chromosomal mutation

Change in number and structure of chromosome

Number

- euploidy- change in whole set of chromosome seen on plants
- Aneuploidy- change in chromosome number in set

gain or loss of a chromosome(s)

Structure

- chromosomal aberration- Common in cancer
  Deletion
  Insertion

Gene mutation

- point mutation= change around single base pair
- Example- sickle cell anaemia
- Deletion and insertion cause frame shift mutation

Mutagen -mutation inducing factor

- Chemical= tobacco
- Physical= UV rays

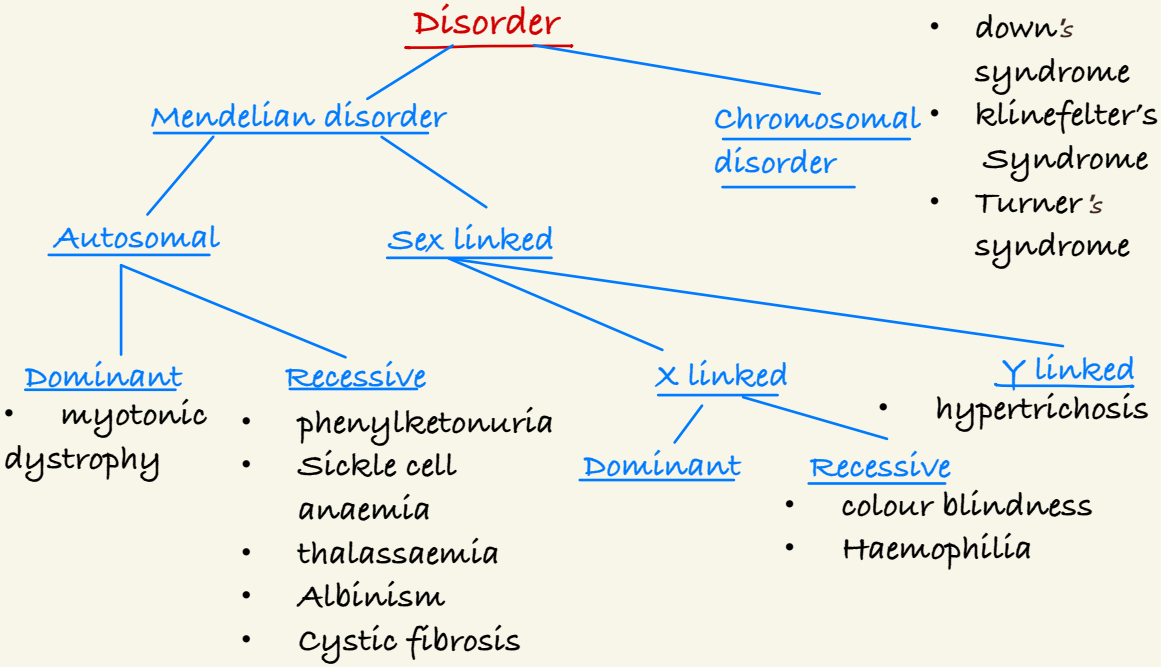
4.7 Mutation

Mutation is a phenomenon which results in alteration of DNA sequences and consequently results in changes in the genotype and the phenotype of an organism. In addition to recombination, mutation is another phenomenon that leads to variation in DNA.

As you will learn in Chapter 5, one DNA helix runs continuously from one end to the other in each chromatid, in a highly supercoiled form. Therefore loss (deletions) or gain (insertion/duplication) of a segment of DNA, result in alteration in chromosomes. Since genes are known to be located on chromosomes, alteration in chromosomes results in abnormalities or aberrations. Chromosomal aberrations are commonly observed in cancer cells.

In addition to the above, mutation also arise due to change in a single base pair of DNA. This is known as point mutation. A classical example of such a mutation is sickle cell anemia. Deletions and insertions of base pairs of DNA, causes frame-shift mutations (see Chapter 5).

The mechanism of mutation is beyond the scope of this discussion, at this level. However, there are many chemical and physical factors that induce mutations. These are referred to as mutagens. UV radiations can cause mutations in organisms – it is a mutagen.



X linked

Dominant

X+. Dominant (disorder)

Xa Recessive

X+ X+ = diseased
X + Xa = diseased
X a Xa = normal
X +Y = diseased
Xa Y = normal

Recessive

X+. Dominant

Xa Recessive (Disorder)

X+ X+ = normal
X + Xa = normal, carrier
X a Xa = diseased
X +Y = normal
Xa Y = diseased

Autosomal

Dominant

A= dominant (disorder)

a= recessive

AA= diseased
Aa= diseased
aa= unaffected

Recessive

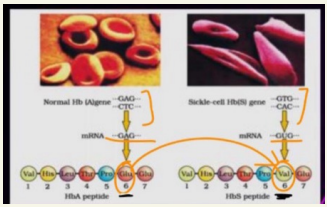
A= dominant

a= recessive (Disorder)

AA= unaffected
Aa= unaffected
aa= diseased

Phenylketonuria





Autosomal recessive
Gene present on chromosome 11

Alleles
- Hbs- affected
- HbA- normal

RBC shape
Biconcave. Sickle

Sickle cell anaemia

- HbA HbA - normal
- HbA HbS - normal / carrier
- Hbs HbS - affected

Hb- polymerisation under low oxygen tension

- Due to point mutation
- GAG change to GUG
- Substitution of glutamic acid by valine in beta globulin chain at sixth position



Quantitative disorder= thalassaemia (autosomal recessive)

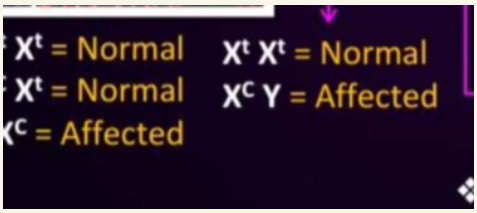
Alpha thalassaemia

- gene present on chromosome 16
- gene = HbA1 and HbA2
- Mutation or deletion of one or more gene cause absence of alpha chain or reduce synthesis
- due to precipitation of beta chain, abnormal haemoglobin is formed and cause RBC to rupture

Beta thalassaemia

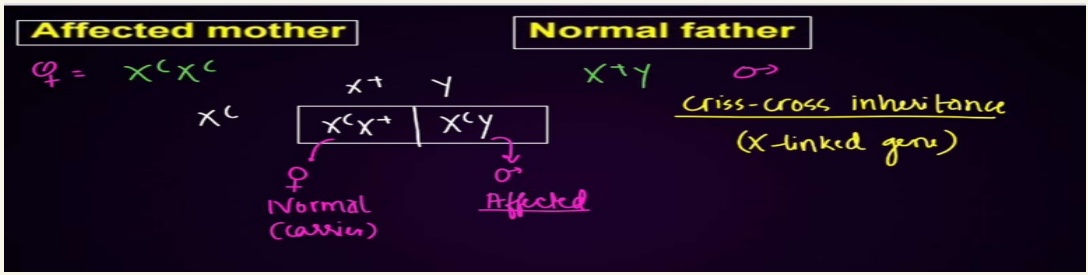
- Gene present on chromosome 11
- gene= HBB
- mutation or deletion of one or both gene cause absence of beta chain or synthesis
- Due to precipitation of alpha chain, abnormal haemoglobin is formed, which cause rupture of RBC

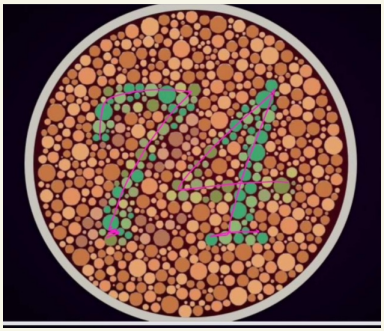
- ultimately, the rupture of RBC cause anaemia
- as there is reduction in number of RBC. This disorder is called quantitative disorder as quantity is affected.

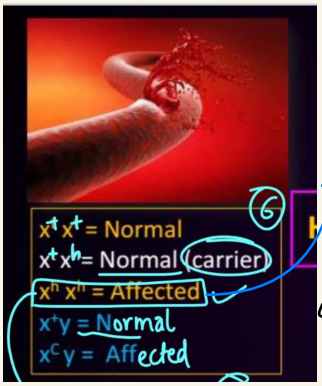


Colour blindness

- Defect in red and green cone of eye
- Unable to discriminate between red and green colour
- Due to mutation in certain genes present on X chromosome
- Walks in about 8% of males and 0.4% of females as male have one X chromosome and female has 2







Death at embryonic stage

Single protein that is part of cascade of proteins, involved in blood clotting is affected

Due to this simple cut will result in non-stop bleeding

Extremely rare in females

Haemophilia

Queen Victoria was carrier of this disease

The family Pedigree of Queen Victoria shows many descendants

Types of haemophilia

- Haemophilia A (Factor VIII)
- Haemophilia B (Factor IX)
- Haemophilia C (Factor XI)

Chromosomal disorder

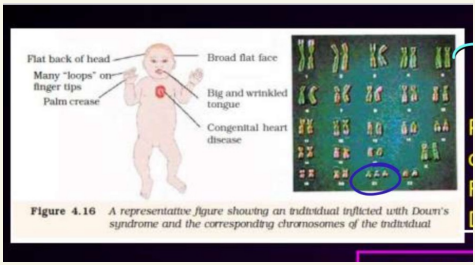


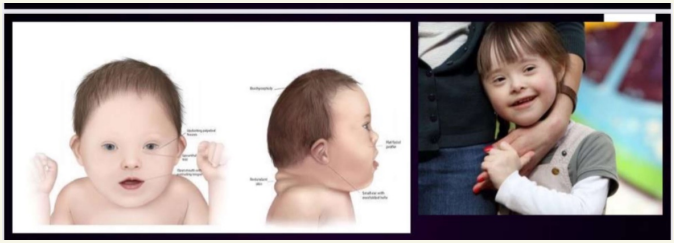
Figure 4.16 *A representative figure showing an individual inflicted with Down's syndrome and the corresponding chromosomes of the individual*

Presence of additional copy of 21st chromosome (trisomy). First described by Longdown down in 1866.

Down syndrome

Symptoms

- short stature
- Small round head
- furrowed tongue
- Partially open mouth
- Broad palm
- Physical psychomotor, and mental development is retarded



A person with trisomy of 21st chromosome shows 1

(i) Furrowed tongue
(ii) Characteristic palm crease
(iii) Rudimentary ovaries
(iv) Gynaecomastia

Select the correct option, from the choices given below :

(A) (ii) and (iv)
(B) (i), (ii) and (iv)
(C) (ii) and (iii)
(D) (i) and (ii)

Klinefelter's syndrome

- additional copy of X chromosome
- Karyotype= 47 or 44+ xxy
- overall: masculine development
- Feminine development is also expressed
- Breast development= gynaecomastia
- Sterile

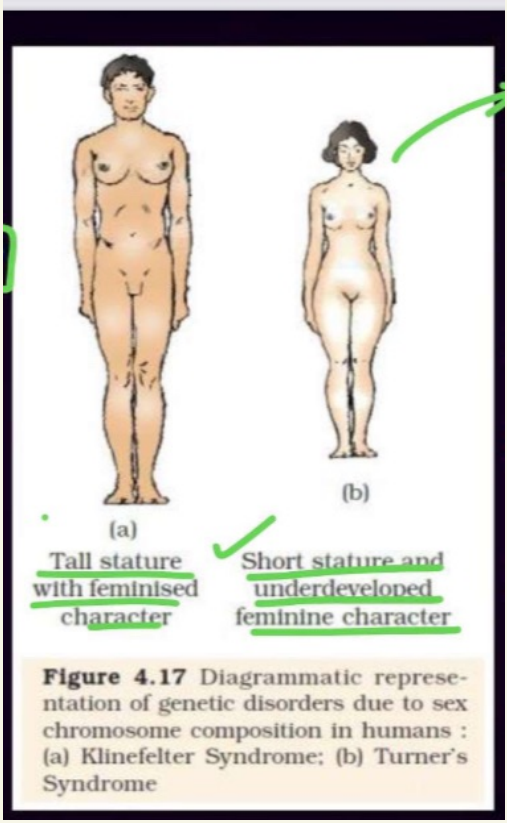
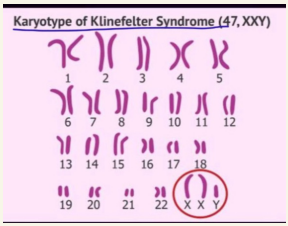
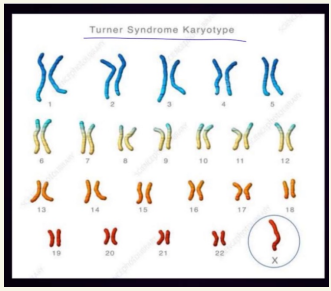


Figure 4.17 Diagrammatic representation of genetic disorders due to sex chromosome composition in humans : (a) Klinefelter Syndrome; (b) Turner's Syndrome

Turner's syndrome

- monosomy due to absence of one of the X chromosome
- 45 with XO
- Sterile female
- Rudimentary ovaries
- Lack of other secondary sexual character





If both the parents are carriers for thalassaemia, the chances of an afflicted child to be born to them is : 1
2024

(A) 25%
(B) 50%
(C) 75%
(D) 100%

Pedigree analysis

- study of family history about inheritance of particular trait for several generation
- Symbols used in human Pedigree analysis

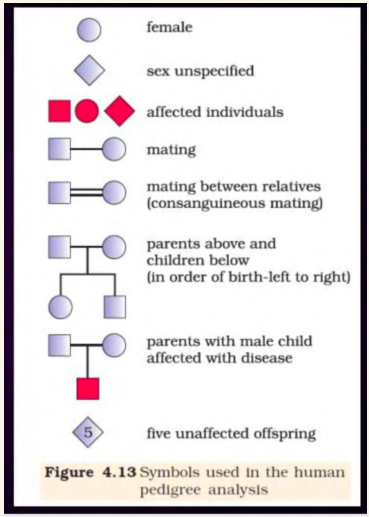


Figure 4.13 Symbols used in the human pedigree analysis

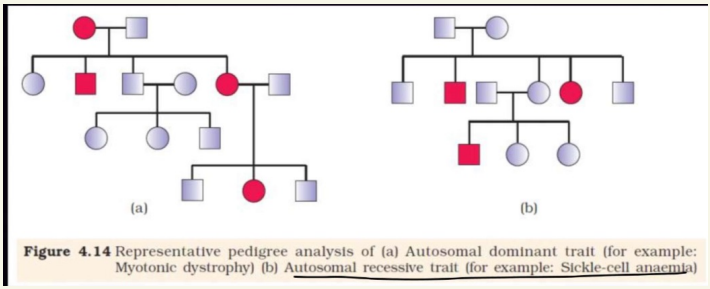
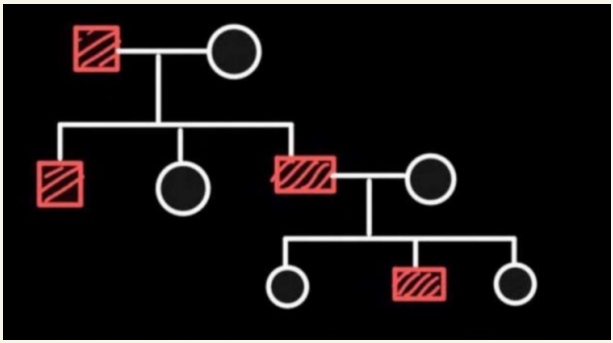


Figure 4.14 Representative pedigree analysis of (a) Autosomal dominant trait (for example: Myotonic dystrophy) (b) Autosomal recessive trait (for example: Sickle-cell anaemia)

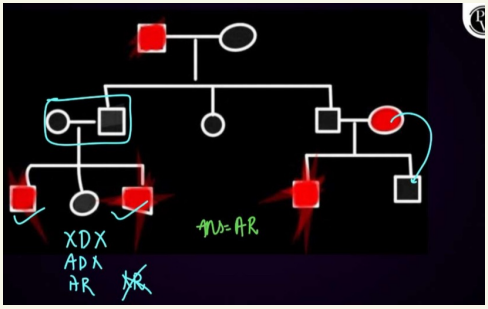
Y linked if father is affected, all son should be affected

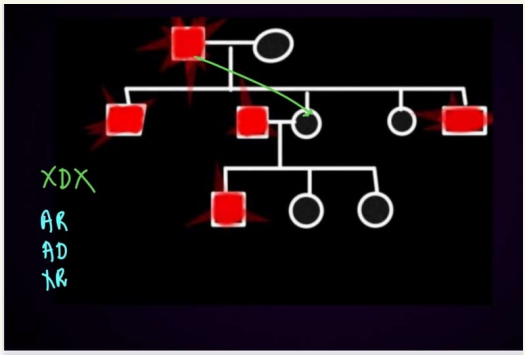


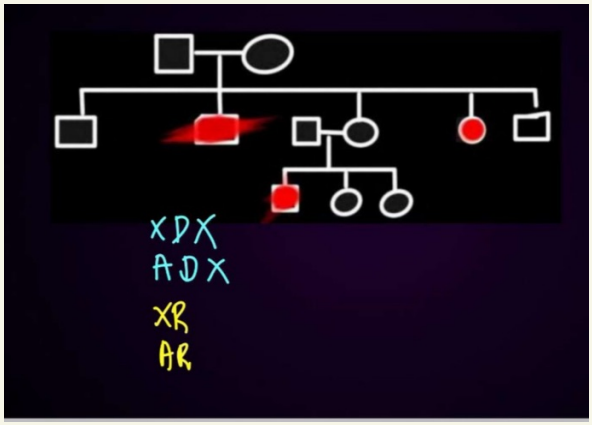
Elimination method

- if both parents are normal and one child is affected , XD and AD not possible
- if both parents are affected and anyone child normal, AR and XR not possible
- XR not possible. ⟶ Father normal daughter affected.
  ↘ Mother affected son normal

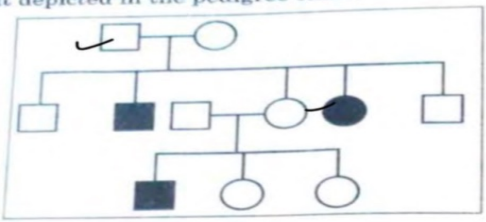
- XD not possible. → Father affected daughter normal.
  → Mother normal and son is affected







Study the pedigree chart of a family showing the inheritance pattern of a certain disorder. Select the option that correctly identifies the nature of the trait depicted in the pedigree chart. 1



NCERT ✓

PYQ.

XDX
AD X

(A) Dominant X – linked
(B) Recessive X – linked
(C) Autosomal dominant
(D) ✓ Autosomal recessive

2024

QUESTION

Among the seven pairs of contrasting traits in pea plant as studied by Mendel, the number of traits related to flower pod and seed respectively were

A 2,2,2

B 1,2,2

C 2,2,1

D 1,1,2

QUESTION

The colour based contrasting traits in seven contrasting pairs, studied by Mendel in pea plant were

Seed
Flower
Pod

A 1

B 2

C 3

D 4

QUESTION

_____ pairs of contrasting traits were studied by Mendel in pea plant.

A 6

B 7

C 8

D 10

QUESTION

Some of the dominant traits studied by Mendel were

A round seed shape, green seed colour and axial flower position

B terminal flower position, green pod colour and inflated pod shape

C violet flower colour, green pod colour, round seed shape

D wrinkled seed shape, yellow pod colour, and axial flower position.

QUESTION

A recessive allele is expressed in

A. heterozygous condition only
B. homozygous condition only
C. F_3 generation
D. both homozygous and heterozygous conditions.

QUESTION

The characters which appear in the first filial generation are called

A. recessive characters
B. dominant characters
C. holandric characters
D. lethal characters.

QUESTION

What will be the distribution of phenotypic features in the first filial generation after a cross between a homozygous female and a heterozygous male for a single locus?

A. 3 : 1
B. 1 : 2 : 1
C. 1 : 1
D. None of these

TT × Tt

	T	t
T	TT	Tt

All tall

QUESTION

In a monohybrid cross between two heterozygous individuals, percentage of pure homozygous individuals obtained in F_1 generation will be

A. 25 %
B. 50 %
C. 75 %
D. 100 %.

Tall Dwarf.

Tt × Tt

↓

TT Tt tt

1 2 1

50%.

QUESTION

Mendel selected how many true breeding pea plant?

A 12

B 13

C 14 ✓

D 15

QUESTION

Which of the following crosses will give tall and dwarf plants in same proportions?

Tall Dwarf

A TT × tt — All tall.

B TT × Tt

T | T | t
T | TT | Tt
All Tall

C tt × tt ✗

D Tt × tt ✓

t
T | Tt ⇒ Tall
t | tt ⇒ Dwarf